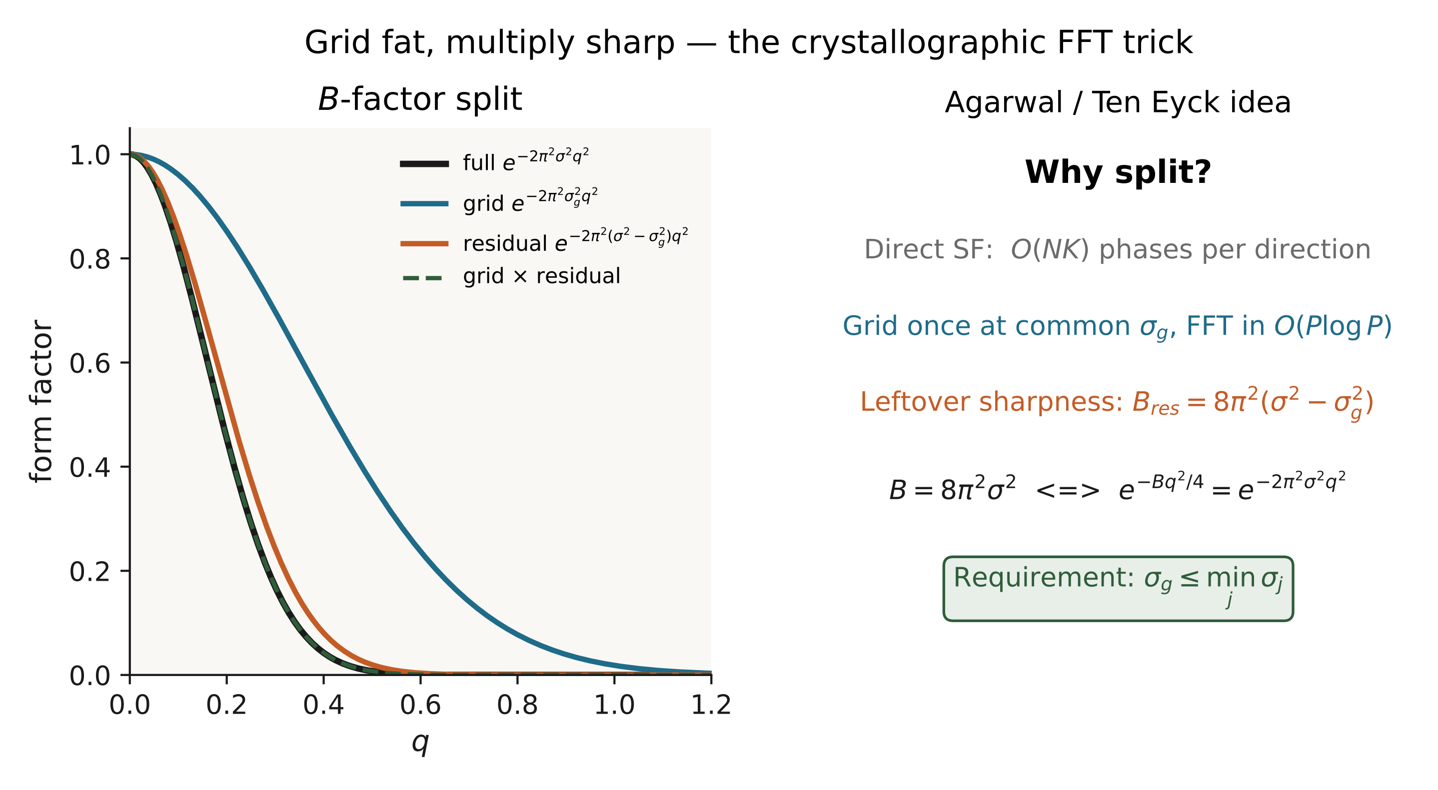
Grid fat, multiply sharp — the crystallographic FFT trick
B-factor split
full $e^{-2\pi^2\sigma^2q^2}$
grid $e^{-2\pi^2\sigma_g^2q^2}$
residual $e^{-2\pi^2(\sigma^2-\sigma_g^2)q^2}$
grid × residual
form factor
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
q
Agarwal / Ten Eyck idea
Why split?
Direct SF: $O(NK)$ phases per direction
Grid once at common σ_g, FFT in $O(P\log P)$
Leftover sharpness: $B_{res} = 8\pi^2(\sigma^2 - \sigma_g^2)$
$B = 8\pi^2\sigma^2$ <=> $e^{-Bq^2/4} = e^{-2\pi^2\sigma^2q^2}$
Requirement: $\sigma_g \leq \min_j \sigma_j$